RAGHAVENDER REDDY BADUGULA

MSc Advanced Mechanical Engineering (Cranfield University) | Simulation (CFD/FEA), Materials & Technical Project Engineering

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PROFESSIONAL SUMMARY

Mechanical, Simulation & Materials Engineering graduate (MSc, Cranfield University) with hands-on expertise in Wire Arc Additive Manufacturing (WAAM) high-temperature molten salt corrosion, Abaqus FEA structural buckling, ANSYS ICEM CFD & Fluent aerodynamic optimization, MATLAB/XFOIL Blade Element Momentum (BEM) & offshore wave mechanics, and engineering project management. Qualified in GATE 2022 (Smart Manufacturing) with 3+ years of cross-functional project coordination and UK team leadership. Combines rigorous physical engineering analysis with AI-augmented systems architecture—designing domain logic and orchestrating AI-assisted software workflows to build end-to-end engineering and localization tools.

CORE TECHNICAL & MANAGEMENT SKILLS

- **CFD, FEA & Numerical Simulation:** Abaqus FEA (3D Shell Partitioning, Linear Perturbation / Eigenvalue Buckling), ANSYS Fluent (k-ω SST Turbulence Modeling), ANSYS ICEM CFD (Multi-Block Structured O/C-Grid Meshing, y+ Boundary Layer Sizing, Grid Independence), XFOIL, MATLAB (Blade Element Momentum [BEM] Solver, Airy Linear/Non-Linear Wave Kinematics), SolidWorks (3D CAD & Kinematics), AutoCAD.
- **Materials, Corrosion & Metrology:** Wire Arc Additive Manufacturing (WAAM SS316L), High-Temperature Molten Salt Corrosion (600°C Ternary $NaNO_3$–KNO_3–KCl), 3D Optical Surface Profilometry / Dimensional Metrology (30-Position Stitched ZVI Interferometry, Cumulative Probability Analysis), SEM-EDS Elemental Mapping & Point-ID, Metallographic Sample Prep, Precision Gravimetry (10^{-5} g), Fracture Mechanics & Structural Integrity (FAD Kr/Lr, Fatigue Crack Growth), COSHH Safety Compliance.
- **Engineering Design & Project Management:** Project Charter & Scope Definition, Double Diamond Design Thinking, Quantitative PM Risk Assessment Matrices (Probability-Impact Scoring & Mitigation), Gantt Scheduling, Circular Economy & UK/ISO Standards Compliance (BS EN 12464-1, ISO 3382-3, BS EN 1729-1, BS 8300), Rapid 3D Printing (Bambu Studio PLA) & Mechatronic Prototyping.
- **Product Logic & AI-Augmented Engineering:** End-to-End System Architecture & Logic Design, AI-Assisted Software Engineering & Execution (Python, FastAPI, Playwright, Git/GitHub Actions CI/CD), Automated Testing Workflows, Local & Cloud LLM Pipeline Integration.

EDUCATION & QUALIFICATIONS

MSc Advanced Mechanical Engineering | Cranfield University, UK — Oct 2024 – Sep 2025

- Key Modules: Fluid Mechanics & Loading (74%), Component Design (73%), Structural Integrity (69%), Industrial Group Project (66%), Principles of Engineering (64%), Engineering Design & Project Management (63%), MSc Thesis (62%), CFD for Renewable Energy (60%), Engineering Stress Analysis (FEA), Assessing Risk & Failure.

B.Tech Mechanical Engineering (Mechatronics) | MGIT, Hyderabad, India — Aug 2016 – Nov 2020

- GATE 2022 Qualified: Smart Manufacturing & Advanced Production Systems. Focus: Mechatronics, CAD/CAM, Kinematics, Thermo-Fluid Systems.

ENGINEERING, SIMULATION & DESIGN PROJECTS (CRANFIELD UNIVERSITY & MGIT)

Molten Salt Corrosion of WAAM SS316L for CSP Thermal Energy Storage (MSc Thesis) | Cranfield University — 2024 – 2025

- Investigated high-temperature corrosion of Wire Arc Additively Manufactured (WAAM) SS316L in ternary chloride molten salt ($10NaNO_3$–$78KNO_3$–12KCl) at 600°C across 250 h and 500 h cycles under open vs. covered-lid crucible conditions to simulate vapour-retentive CSP tank zones.
- Performed 30-position stitched 3D optical dimensional metrology (.zvi surface profilometry & cumulative probability radius-change distributions), precision mass metrology (10^{-5} g), metallographic sectioning/mounting/polishing, and cross-sectional SEM-EDS elemental mapping.
- Demonstrated that covered-lid vapour retention accelerates porous scale formation, Na/K/Cl/O ingress, and early pit initiation at 250 h compared to compact Fe/Cr-rich oxide growth in open crucibles, informing CSP TES venting and alloy selection.

Fluid Mechanics & Offshore Loading: HAWT BEM Rotor & Airy Wave Kinematics | Cranfield University (74%) — Jan 2025

- Wind Turbine Rotor Aerodynamics (Part A): Generated NACA 0030 airfoil lift/drag polars in XFOIL and formulated an iterative Blade Element Momentum (BEM) convergence algorithm in MATLAB for a 3-bladed HAWT rotor (R = 7 m), computing radial induction factors, total thrust (1,804.6 N), torque (1,693.8 Nm), power (10.16 kW), and Cp (0.1872), alongside blade-twist optimization.
- Offshore Hydrodynamic Wave Loading (Part B): Applied Linear (Airy) and Non-Linear wave theory to derive horizontal and vertical fluid velocity potentials, simulating depth-dependent velocity profiles and elliptical particle trajectories (z = 0 to -2 m) in MATLAB to evaluate wave forces on offshore monopile met masts.

FEA Buckling & Stress Analysis of Offshore Wind Turbine Monopile Foundations | Cranfield University — Abaqus FEA

- Modeled a 40 m offshore wind turbine hollow steel monopile (R = 3.75 m, t = 0.08 m) in Abaqus using 3D shell elements partitioned via horizontal and 90° circumferential datum planes with encastre seabed boundary conditions.
- Constructed a 23,200-element structured rectangular mesh (0.2 global seed) to eliminate tetrahedral noise and executed Linear Perturbation (Eigenvalue Buckling) analysis under combined eccentric turbine/nacelle weight, marine wave pressure (1.5 kPa), and sudden seabed pressure (20.3 kPa).
- Extracted the first 5 critical buckling modes (eigenvalues λ_1 = 0.3349 to λ_5 = 0.5917), characterizing progressive multi-wave shell instabilities and fixed-base stress concentrations against DNV offshore structural standards.

ICEM CFD Aerodynamic Simulation & Biomimetic Airfoil Optimization for Urban VAWTs | Cranfield University ANSYS ICEM & Fluent

- Modified a symmetric NACA 0015 airfoil for urban Vertical Axis Wind Turbines (VAWTs) at low-to-moderate wind speeds (4 m/s & 12 m/s) by introducing a 40%-chord upper-surface bump and a stingray-inspired curved trailing edge to delay dynamic stall.
- Built a C-shaped computational domain (-20c to +40c) with multi-block O-grid structured meshing in ANSYS ICEM CFD (near-wall y+ spacing 3.3×10^{-6} m, 2×2×2 determinant quality > 0.837), validated via a 4-level grid independence study (21k to 104k nodes, <1% variation).
- Simulated 2D viscous flows using the k-ω SST turbulence model in ANSYS Fluent, demonstrating prolonged upper-surface flow attachment and increasing lift coefficient (CL) from 0.80 to 0.95 at 10° angle of attack.

Smart & Sustainable Shower System — Component Design & Prototyping | Cranfield University (73%) SolidWorks & 3D Print

- Designed, modeled in SolidWorks, and 3D-printed (Bambu Lab PLA) a closed-loop cold-water recirculation shower prototype integrating temperature sensors, solenoid valves, an under-tub storage reservoir, and a return pump to boiler.
- Conducted stakeholder influence analysis, functional/non-functional systems requirements definition, and physical prototype evaluation targeting a 50–70% reduction in household pre-heating water wastage.

Engineering Design & Project Management: Stafford Cripps SC3 Smart Classroom Modernization | Cranfield University Project Lead

- Authored the Project Charter, Double Diamond design-thinking justification, circular-economy furniture reuse strategy, and UK/ISO compliance plan (BS EN 12464-1 LED lighting, ISO 3382-3 acoustics, BS EN 1729-1 ergonomics, BS 8300 accessibility).
- Built a 15-item quantitative PM Risk Assessment Framework (mitigating high-risk PI scores from 25 to ≤9), Quality Assessment Matrix, and multi-phase Gantt schedule across Initiation, Planning, Execution, and Evaluation.

Lead-Free Semiconductor Solar Paint Formulation (with Nissan) & Marine Coating Testing | Cranfield & MGIT 2024 – 2025

- Nissan Group Project: Formulated a photovoltaic paint matrix dispersing bismuth methyl iodide quantum dots (1.8–2.3 eV) within a TiO_2 electron transport layer on galvanized steel substrates; conducted 48-hour electrical conductivity trials and led technical report synthesis.
- Marine Corrosion & B.Tech Mechatronics: Executed hypothesis testing proving double-layer hull coatings optimize anti-corrosion efficiency; designed a 3D SolidWorks Multi-Terrain Quadruped Robot and conducted thermoacoustic engine simulations at MGIT.

PRODUCT ARCHITECTURE & AI-AUGMENTED SOFTWARE SYSTEMS

Polyglot Engine — Autonomous Multilingual UI Layout Scanner & Repair CLI | github.com/Raghavender1234/polyglot-engine 2026

- Product Logic & Architecture: Conceived and architected a 6-layer multilingual UI verification system connecting browser DOM geometry (Playwright), OpenType shaping (HarfBuzz), and Unicode BiDi isolation to diagnose and repair translation overflow and CSS layout defects.
- AI-Augmented Execution: Designed the end-to-end diagnostic/backtracking repair logic and orchestrated AI-assisted coding workflows to build, test (pytest + 50-case visual benchmark), and package a turnkey CLI and GitHub Action.

IndiCAT (indicat.co.in) & Operations Automation Tools (UK Rota Optimizer, AI LMS) 2024 – Present

- Conceived the product workflows and system architecture for IndiCAT (Windows desktop CAT & subtitle localization app with local/cloud LLMs and WhatsApp/Telegram sync), a UK Food Store Rota Optimizer, and a Multi-Tenant AI LMS—translating real operational pain points into working end-to-end software via AI-assisted development.

WORK & OPERATIONAL LEADERSHIP EXPERIENCE

Team Lead / Operational Specialist | Smashburger, Milton Keynes, UK Nov 2024 – Present

- Promoted to Team Lead; directing shift operations, staff workflows, quality control, health & safety compliance, and inventory management in a high-volume UK hospitality environment.

Assistant Project Manager / Operational Coordinator | Exam & Localization Operations (NTA, TCS iON, IBPS via Vendors) Jun 2021 – Sep 2024

- Managed end-to-end localization, translation, and data operations for high-stakes national entrance exams (JEE Mains, CUET 2023) at the National Testing Agency (NTA), Delhi, plus TCS iON and IBPS banking exams.
- Coordinated cross-functional teams of freelance translators, SMEs, and typists across regional languages, enforcing 100% SLA deadline compliance and strict examination confidentiality protocols.

Industrial Training & Early Career | RINL Vizag Steel Plant, Maruti Suzuki (Varun Motors) & Media Localization 2018 – 2021

- Completed industrial training at RINL Vizag Steel Plant (Coke Plant & I-beam rolling) and Maruti Suzuki; delivered English–Telugu subtitle localization for Netflix/Amazon Prime releases and UK remote interpretation.